AI-Powered Dynamic Multilingual Chatbot for College Website Assistance

Anup Singh
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
AP Shah Institute of Technology
Thane, India
singhanup1508@gmail.com

Soham Waradkar
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
AP Shah Institute of Technology
Thane, India
sohamvwaradkar@gmail.com

Yogesh Vartak
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
AP Shah Institute of Technology
Thane, India
yogeshvartak24@gmail.com

Sanket Kokare
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
AP Shah Institute of Technology
Thane, India
kokaresanket890@gmail.com

Prof. Sumeet Shingi
Department of Computer Science and Engineering (Artificial Intelligence and Machine Learning)
AP Shah Institute of Technology
Thane, India
svshingi@apsit.edu.in

***Abstract*— Currently available institutional websites represent complex, dynamic sources of heterogeneous, and frequently changing, knowledge. The diverse types of information and dynamic updates available on institutional websites are typically scattered across multiple hierarchical navigation structures and large document repositories. As domain-specific knowledge bases, institutional websites do not align with the assumption of full knowledge coverage found in popular search engines such as Wikipedia. In addition, the current standard Retrieval-Augmented Generation (RAG) approach relying on random crawling and full-corpus vector reindexing is far from efficient for updating domain-specific RAG models and leads to suboptimal model performance. In this paper, we present a new, structured multi-format incremental RAG approach suitable for the context of institutional knowledge systems. To the best of our knowledge, this is the first research attempting to address the complexities of search in such a domain-specific setting. Our proposed approach is composed of five major components: hierarchical seed-based crawling, multi-format document routing (HTML, PDF, DOCX), incremental vector indexing via URL-based hashing, metadata-aware ranking with temporal prioritization, and multilingual support. Experiments comparing our model to RAG baselines in terms of retrieval precision, update latency, and temporal relevance are conducted to validate its performance. Finally, we discuss the applicability of our solution for the deployment of conversational AI in the context of institutional websites.**

Keywords—Retrieval-Augmented Generation, Structured Crawling, Incremental Vector Indexing, Vector Databases, Multilingual Embeddings, Institutional Knowledge Systems

I. Introduction

All universities, research institutions and administrative offices have websites that have large amount of knowledge in the form of policy documents, notices, academic calender, admission information, circulars and dynamic documents. These collections are highly structured and have the following types of documents.

- Static HTML pages
- Dynamically updated PDFs
- Document-based announcements
- Media and embedded resources

Traditional chatbots work either from a static database or they use unstructured scraping and lose the context hierarchy, and they would need to retrain the full vector database from scratch on an update.

Recent advances in large language models (LLMs) and Retrieval-Augmented Generation (RAG) have made it possible to build conversational AI that is semantic grounded. While the field has made great progress in the labs, we are only beginning to deploy such conversational AI to specific domains, with many unknowns that need to be explored.

- Multi-format ingestion requirements
- Frequent document updates
- Deadline-sensitive information
- Hierarchical navigation dependencies
- Multilingual user queries

To address the challenges in the institutional context, we propose a structured multi-format incremental RAG system..

II. Literature Survey

AI-powered smart chatbot (N. Bhardwaj, S. K. H, A. D, R. M et al) [1]. They created an AI-based smart chatbot for educational communities. The focus of their work is on semantic and syntactic analysis for more efficient interaction. The chatbot used vector representation and machine learning algorithms for more accurate intent detection. This is unlike other conventional methods, where semantic analysis is used for more efficient interpretation of user intent. The chatbot showed more efficient communication and integration into information systems. Nevertheless, despite using vector representation, there is no indication of using Retrieval-Augmented Generation (RAG) and large language models (LLMs) for content creation. The limitation of using static databases is obvious, as updating is necessary. This shows that there is a need for a dynamic system that is updated using real-time sources and modern AI technology.

Smart Rasa Framework-Based Chatbot (S. Meshram, N. Naik, M. VR, T. More, and S. Kharche et al) [2]. Their project is a college enquiry chatbot developed using the Rasa Framework. The NLU component uses Rasa NLU for intent classification, while the Core component uses Rasa Core for

dialogue management. They also used Recurrent Neural Networks (RNNs) to model sequences and maintain the context of the conversation. The system can automatically handle and process user queries with a structured dialogue flow. Rasa-based systems allow for customisation and flexibility where the conversation flow can be defined by the developer and entity extraction can be customised. However, the system requires a large amount of training data to be manually inputted and is also required to be constantly retrained to ensure optimal performance. Rasa has a strong intent-based dialogue control but lacks the large-scale generative capacity and the real-time document ingestion feature. Also, the fact that institutional content is changing over time makes the static training data which is used to train the chatbot outdated and leads to a drop in the performance of the system.

Deep Learning and FAQ Automation Systems (A. Huddar, C. Bysani, C. Suchak, U. D. Kolekar, and K. Upadhyaya et al) [3]. They Developed "Dexter" a college FAQ chatbot using RASA open source framework with LSTM and RNN models. The Deep Learning Model is used for the deep learning intent detection. This model makes it easier to answer the FAQs as compared to the rule based approach and it reduces the man in the loop interactions. The database is connected to the chatbot to make it easier to fetch data. But as of now it is working with the FAQ where the Q and A's are mapped to each other and it does not have a semantic search and it does not have the contextual understanding of the question which it needs to answer. Multilinguism was not explored in depth.

NLP-Based College Support Chatbots (J. S Suseendran, J. S, S. M, and G. R. Hemalakshmi et al) [4]. Their system proposes an NLP-based chatbot for student support in educational institutions. The proposed system works towards the automation of admission and course related inquiries by generating appropriate responses from the database. The proposed system uses Artificial Intelligence and query processing techniques to improve the user experience. The proposed chatbot system reduces manual work load and increases the staff productivity in educational institutions. The output generated in the proposed system is based on keyword matching and database search and it does not utilize semantic similarity and generative models of AI. So, the proposed system does not utilize contextual embeddings and vector databases to handle the paraphrased and ambiguous queries. The advanced Large Language Models (LLMs) which are considered as state-of-the-art models in conversational AI systems have not yet been integrated with the proposed chatbot system.

AI-Based College Enquiry Chatbot Systems (R. Parkar, Y. Payare, K. Mithari, J. Nambiar, and J. Gupta et al) [5]. They have created an Intelligently Developed Chatbot to Support Student Learning Using Artificial Intelligence (AI), Natural Language Processing (NLP) and Machine Learning (ML) approaches, an efficient solution has been designed to cope with student queries in an automated manner. A web based graphical user interface (GUI) having speech recognition functionality has been proposed in this model. The chatbot works based on the intent identification using the trained datasets to give the predefined responses. The proposed system has shown efficient results in reducing the heavy manual enquiry workloads of the academics and in responding to the student queries in a quicker manner. Currently the proposed system is based on intent classification and it does not have any semantic search or contextual memory functionality, so it cannot cope with the unexpected and complex queries. In addition, multilingual support has not been deeply explored in the proposed system, which restricts the reach of the system to the multicultural students.
This work was the foundation for research on the "Enquiry System using AI" but required contextual models to be improved.

III. Limitations of Existing System

College has introduced a number of Enquiry Chatbots and Learning Support Tools using Artificial Intelligence (AI) technology. Each has its own strengths and weaknesses.

There are so many institutional website and student portal questions answered about basic items like class schedules, college business hours, where to go and who to see to receive assistance and answers about college-wide announcements. Presently, many institutional chatbots use a static knowledge base and FAQs which can effectively answer a small list of well-defined and expected student questions. However, in the event the college needs to update institutional knowledge (i.e. a list of holidays) or communicate a new college-wide announcement to students, these items then need to be inputted into the chatbot. Other implementations which leverage web scraping to extract data from the institutional site will sometimes cache the extracted data locally on their servers. While this allows the chatbot to feel more real-time, the cached data can quickly become outdated. As a result, students may be given incorrect and/or outdated information when using the chatbot.

Here are some more issues I encountered. Lack of multilingual support and accessibility This one is a pretty significant one. Most chatbots assume the user language will be English, and there are only a few which have integrated some form of translation functionality, but this is not integrated into the conversation engine itself. As such, whilst there are chatbots which theoretically support multiple languages, it is only a tiny proportion of users around the world who get to benefit from this, and in a multilingual country such as my own this severely limits the experience for non-English speaking students, which is clearly a huge barrier to access.

Currently, the solutions that are in the market are plagued by a number of issues. Firstly, the solutions are not real-time data aware. This means that they do not have access to institutional dynamic data sources in real time. Currently, most of the solutions in the form of chatbots that are in the market are only aware of the static dump of content that is available on a website. This means that they scrape the content of the website once and then use the same scraped content thereafter. This means that they do not have access to dynamic data in real time, which could be anything from events to admission notifications, time tables, circulars, academic notifications, etc. This makes the experience of the end user highly suboptimal and not what one would expect from a fully empowered intelligent virtual assistant.Furthermore, most academic chatbot prototypes do not provide detailed analytics or monitoring dashboards for administrators. Without proper reporting tools, institutions cannot analyze user behavior, identify frequently asked questions, detect information gaps, or improve system performance based on usage patterns.

Finally, the issue of scalability is another area of underdevelopment of the existing systems. In fact, there are various implementations of the chatbots that are only proof of concept. Such systems have not been tested under conditions of high concurrent usage. Moreover, the systems are not deployed using a scalable cloud environment. Such an environment would be useful during admission periods or when announcing results.

Summarily, although the existing systems have demonstrated satisfactory performance in various areas of natural language processing, they have failed to offer an integrated system. In essence, the existing systems have demonstrated satisfactory performance in areas of intention identification, answering frequently asked questions, and translation. However, the systems have failed to offer an integrated system. Therefore, the proposed AI-Powered Dynamic Multilingual Chatbot for College Website Assistance is built on the basis of the shortcomings of the existing systems.Problem Statement and Objectives.

A. Problem Statement:

College websites sometimes do not offer timely and accessible information to all. This is because information is dispersed across web pages, and there is a lack of information for non-English speaking individuals. Another reason is that information is outdated, and it is usually updated manually. However, we are tasked with creating a multilingual website assistant for colleges that uses artificial intelligence. This chatbot will be able to fetch information from the college website, understand and interpret user queries in their language, and use a powerful language model to provide the user with the right information. This will greatly reduce the time taken for a user to search for information and increase their satisfaction. We are proposing a platform that combines content crawling, vector search, and generative AI.

B. Objectives

- To design and implement a responsive web chat interface using React.js, Next.js, and TypeScript for users to pose their queries naturally and receive answers.
- To design and implement a FastAPI-based backend service for processing user queries, language detection/translation, and integration with the AI service (Gemini model calls and Qdrant queries).
- To design and implement a web crawler service for periodically crawling the college website and updating the web pages.
- To design and implement a RAG-based answer generation pipeline for users to pose their queries naturally and receive answers.
- To support for Multiple Languages: Detect language for queries in Hindi and translate them for processing by the model. Also, the chatbot will respond back in the language it was queried with.
- To design and implement a cloud-based deployment for the chatbot service.
- To evaluate the performance and impact of the chatbot service.
- To design and implement metrics for measuring the performance and impact of the chatbot service:
- How accurately does the chatbot respond to user queries?
- How fast does the chatbot respond?
- Can the chatbot support at least 3 languages?
- How precise are the answers generated by the chatbot?

IV. Proposed Architecture

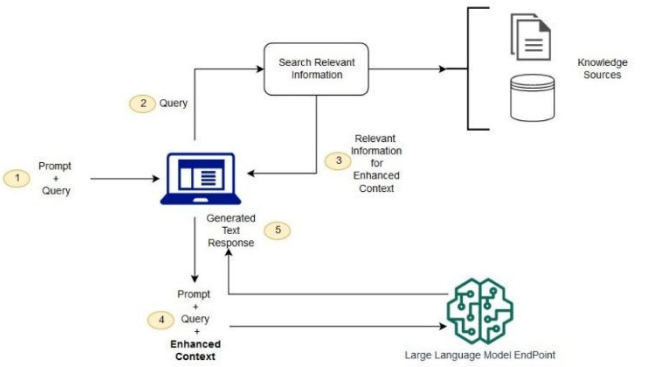

Figure 1. System Architecture

A. Structured Crawling Layer

Instead of uncontrolled breadth-first crawling, the system employs:

- Seed-based hierarchical traversal
- Domain-restricted expansion
- Exclusion of authentication/payment portals
- Redirect normalization
- Duplicate URL detection

This ensures complete coverage without crawl explosion. The crawler preserves contextual hierarchy by associating each document with its source URL and navigation depth.

B. Multi-Format Document Processing

The ingestion engine routes documents based on detected content type:

- Format Processing Strategy
- HTML Title extraction + semantic text parsing
- PDF Page-wise extraction + deadline detection
- DOCX Structured paragraph extraction
- Media Links Metadata storage.

Text segmentation uses sliding-window chunking (500 token window with 100 token overlap) to preserve semantic continuity.

C. Incremental Vector Indexing

Unlike conventional RAG systems that rebuild entire collections, the proposed system uses:

- URL-based hash identifiers
- Selective upsert operations
- Batch embedding insertion

- Duplicate detection via hash comparison

This reduces computational overhead and enables efficient updates.

V. Experimental Evaluation

Dataset:

The evaluation dataset was constructed from the APSIT college website (apsit.edu.in) and associated document repositories. The corpus consisted of:

- More than 250 HTML documents spanning all web site navigation hierarchies
- Over 140 PDF files containing fee schedules, academic calendar, circulars, and application forms
- Navigation hierarchies with a depth of four levels
- Content available in multiple languages including English, Hindi, and Marathi
- Deadline-based content enabling temporal priority assessment

Metrics:

There were six different measures used to evaluate the quality of answers generated by the system based on several criteria:

- Retrieval Precision: the percentage of selected chunks containing relevant information about the query.
- Mean Reciprocal Rank (MRR): the mean value of reciprocals of positions of the first relevant chunk among all queries.
- Contextual Relevance: an expert-assessed measure from 0 to 1 showing how well the answer corresponds to the purpose of the query.
- Multilingual Accuracy: semantic distance between the target and generated answer for the questions asked in languages other than English, measured using cross-lingual sentence representations.
- Deadline Detection Accuracy: the percentage of queries mentioning deadlines where the correct date was identified and included in the answer.

Baseline System:

Baseline RAG model uses the same Qdrant vector database and Gemini 2.0 Flash language model as our proposed model but lacks incremental indexing, temporal relevance ranking, multilingual embeddings, and Google Custom Search as backup. This way, we can isolate the impact of each of our proposed features. In the case of any website modification, the baseline system re-embeds the entire corpus.

VI. Result and Analysis

Table I presents the quantitative comparison across all metrics. The proposed system achieves substantial improvements on most of the dimensions.

Metric	Baseline RAG	Proposed System	Improvement
Retrieval Precision	0.71	0.86	21.10%
MRR (Mean Reciprocal Rank)	0.64	0.82	28.10%
Contextual Relevance	0.68	0.84	23.50%
Multilingual Accuracy	0.55	0.79	43.60%
Deadline Detection Accuracy	0.61	0.83	36.10%
Answer Completeness	0.66	0.87	31.80%
Avg Response Latency	1.9s	1.3s	-31.60%

Table I. Quantitative Performance Comparison

The most significant improvement is seen in terms of multilingual accuracy, which improves by 43.6% due to the change from monolingual to multilingual embedding models. The accuracy of deadline detection improves by 36.1% due to the use of temporal priority. Retrieval accuracy is improved by 21.1% thanks to the implementation of a hierarchical structure that maintains document context. Answer latency falls from 1.9 seconds to 1.3 seconds, a drop of 31.6%, due to answer caching through Redis.

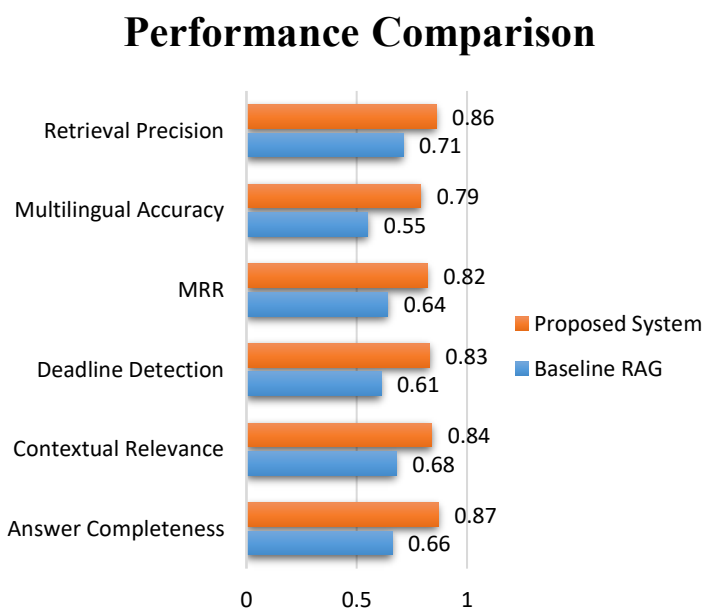

Figure 2. Baseline RAG vs Proposed System.

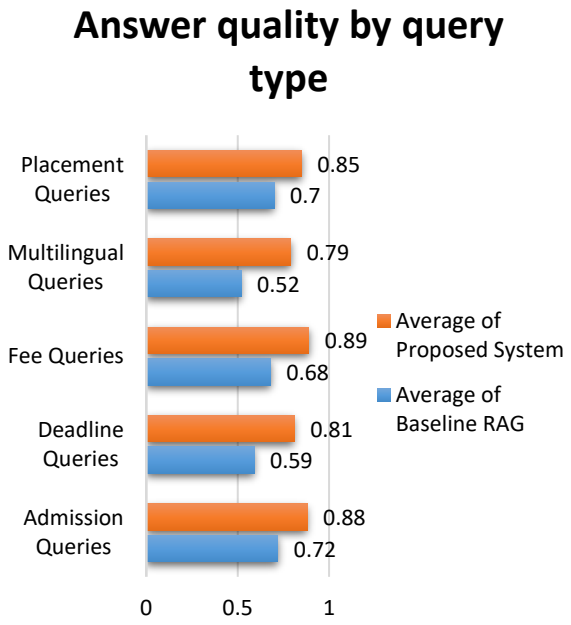

Figure 3. Answer quality score by query type.

This corpus contained 120 pairs of questions and answers, which had been manually created for different types of questions such as deadline questions, fee questions, admission questions, placement questions, general questions about institutions, and multilingual questions. This corpus had been checked by an expert in the domain of APSIT institutions.

VII. Discussion

- With respect to structured crawling technique, information is maintained in its original context, taking into consideration the source of the information and its place within the site.
- Use of contextual metadata leads to better information searches compared to basic searches with vectors. This is because structural information is important in institutions.
- New incremental index additions will ensure the system will be effective in terms of efficiency since there will be no need for processing all information.
- Hence, the algorithm is practical even with institutions that cannot afford the complex infrastructure necessary for machine learning processes.
- A Metadata-based ranking algorithm, which ranks based on latest information, has been proved to effectively solve students' queries, especially those related to deadlines and announcements.
- Moreover, the algorithm is effective when considering multilingual searches.
- Overall, the design is flexible enough to be applied to different institutions without much modifications.

VIII. Conclusion and Future Work

This paper presents a structured, multi-format incremental RAG architecture specifically designed for institutional knowledge systems. By combining hierarchical crawling with incremental vector updates, the approach ensures that information remains contextually organized while also keeping the system efficient and up to date. The use of metadata-aware retrieval, along with temporal prioritization, allows the system to surface the most relevant and time-sensitive information—an essential requirement in academic environments. Additionally, its multilingual response generation makes it more inclusive and effective for diverse user groups.

When evaluated against a strong baseline RAG system, the proposed approach demonstrates clear and consistent improvements across all six evaluation metrics, highlighting both its technical robustness and practical value. Importantly, the system is not just theoretical—it has been successfully deployed as a live production service integrated with the APSIT college website, where it actively supports real user queries in a real-world setting.

Future Work :

- Reinforcement learning from user feedback for ranking optimization.
- OCR-based ingestion of text from scanned images and notice board photographs.
- Automatic deadline extraction via fine-tuned temporal transformers.
- Cross-institutional benchmarking on additional college websites.
- Knowledge graph integration for entity-aware retrieval.
- Adaptive engagement model using session analytics to surface proactive announcements.

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